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AI Learning

Page 1: Git & GitHub Strategic Workflow

1. Identity & Environment Setup

Setting up Git is the first step to ensure every code commit is saved under your name.

- **Set User Name:** git config --global user. name "[Your Name]" — Identifies you as the author of the code.
- **Set User Email:** git config --global user.email "[Your Email]" — Links your work to your professional email.
- **Initialize Repository:** git init — Used to turn a local folder into a new Git repository.
- **Clone Repository:** git clone [url] — Used to download existing projects from the internet to your computer.

2. The Development Loop (Stage & Snapshot)

These commands track your daily logic and any changes you make to the code.

- **Check Status:** git status — Shows which files have been modified or are ready to be saved.
- **Stage Changes:** git add [file] — Prepares specific files to be included in your next save (commit).
- **Commit Progress:** git commit -m "[message]" — Saves your staged work as a permanent snapshot in history.
- **View Differences:** git diff — Shows exactly what changed in your files but hasn't been staged yet.
- **Unstage File:** git reset [file] — Moves a file out of the staging area if you added it by mistake.

Advanced Collaboration & Safety:

3. Branching Strategy (Feature Development)

In GenAI projects, branches are the best way to work on different prompts or features separately.

- **List Branches:** `git branch` — Shows all the branches you have created.
- **Create Branch:** `git branch [branch-name]` — Creates a new branch so you can work on a specific feature safely.
- **Switch Context:** `git checkout [branch-name]` — Lets you create and switch between different branches.
- **Merge Work :** `git merge [branch]` — Combines the history of a finished branch into your current one.
- **Audit History :** `git log` — Shows the full list of all previous commits in your project.

4. Remote Sync (Sharing & Cloud Backup)

Keep your local work synced with the GitHub cloud for backup and sharing.

- **Add Remote :** `git remote add [alias] [url]` — Connects your local project to an online GitHub link.
- **Push Code:** `git push [alias] [branch]` — Uploads your local work to your online GitHub repository.
- **Pull Updates:** `git pull` — Downloads and merges any new changes from the online repo to your computer.
- **Fetch Changes:** `git fetch [alias]` — Checks for updates from the online repo without changing your local code.

5. Temporary Storage & Cleanup

- **Stash Changes:** `git stash` — Temporarily hides your unfinished work so you can switch branches quickly.
- **Pop Stash:** `git stash pop` — Brings back your hidden work so you can continue where you left off.
- **Delete File :** `git rm [file]` — Removes a file from the project and stages that change.